Ampere SDV End-to-End Traceability

LookML Semantic Layer: Design Assumptions & Specification Checklist

Date: October 30, 2026 | Version: 1.0 (For Customer Validation)

1. Executive Overview

Based on the provided data model diagram (`input_mermaid`) and physical BigQuery schemas (`gold_tables_schema.csv` and `raw_tables_schema.csv`), an initial LookML semantic layer has been generated for the **Ampere Software-Defined Vehicle (SDV) Traceability** project.

To avoid ambiguity between the data warehouse and Looker, every LookML view is named **identically to its underlying physical table**, and only the 8 tables that directly feed the LookML layer are modeled:

Physical Table Name	Layer / Source	Mermaid Node	LookML View Name
`gold_dim_req`	Gold Dimension	`Requirement_View`	`gold_dim_req` (48 cols)
`gold_dim_ts`	Gold Dimension	`TestSet_View`	`gold_dim_ts` (41 cols)
`gold_dim_fact_req_tc`	Gold Fact Bridge	`FACT_Requirement_TestCase`	`gold_dim_fact_req_tc` (11 cols)
`FACT_TC_TR_TS_REQ`	Gold Fact Bridge	`FACT_TC_TR_TS_REQ`	`fact_tc_tr_ts_req` (5 cols)
`FACT_TestRun_Issue`	Gold Fact Bridge	`FACT_TestRun_Issue`	`fact_testrun_issue` (2 cols)
`TBL_CB_SDV_Raw_TC_V2`	Silver / Raw	`TestCase_View`	`tbl_cb_sdv_raw_tc_v2` (114 cols)
`TBL_CB_SDV_Raw_TR_V2`	Silver / Raw	`TestRun_View`	`tbl_cb_sdv_raw_tr_v2` (114 cols)
`TBL_CB_SDV_Raw_Issue_V2`	Silver / Raw	`Issue_View`	`tbl_cb_sdv_raw_issue_v2` (18 cols)

Note: Upstream Silver tables (`TBL_CB_SDV_Raw_Req_V2`, `TBL_CB_SDV_Raw_TS_V2`, `TBL_CB_SDV_Raw_ISP_V2`, and `TBL_CB_SDV_Raw_Initiative_V2`) are already consolidated into `gold_dim_req` and `gold_dim_ts` and are therefore excluded from LookML.

2. Inferred Elements Requiring Customer Validation

While the table schemas and high-level lineage were provided, six technical and business areas had to be **inferred** to build a complete, runnable LookML project. Please review and confirm or adjust the assumptions below.

2.1 Missing Schema for `FACT_TestRun_Issue`

- **Observation:** `FACT_TestRun_Issue` is present in the architecture diagram (bridging Test Runs and Issues), but its DDL/schema was not included in `gold_tables_schema.csv`.
- **Current Assumption:** The physical table was assumed to be named `FACT_TestRun_Issue` with two `NUMERIC` columns: `TR_Id` and `Issue_Id`.
- **Information Needed:** Exact BigQuery table name (e.g., `FACT_TestRun_Issue` vs. `gold_dim_fact_tr_issue`) and exact column names/data types.

2.2 Explore Join Topology, Base View & Cardinalities (`traceability`)

- **Observation:** The diagram connects all 5 entity views and 3 bridge tables into `Traceability_Explore`, but does not specify join types (`LEFT OUTER` vs. `INNER` vs. `FULL OUTER`), relationship cardinalities, or exact `ON` predicates.
- **Current Assumption:**
  1. **Base View:** `gold_dim_req` is the primary starting view of the Explore.
  2. **Req ↔ TC Bridge:** `gold_dim_req` → (`LEFT OUTER`, 1:N) → `gold_dim_fact_req_tc` → (`FULL OUTER`, N:1) → `tbl_cb_sdv_raw_tc_v2` (using `FULL OUTER` so orphan Test Cases without a linked Requirement remain queryable).
  3. **TC/TR/TS/Req Bridge:** `fact_tc_tr_ts_req` is joined on `tc_id = tbl_cb_sdv_raw_tc_v2.id`

`AND (req_id = gold_dim_req.req_id OR req_id IS NULL)`, then `LEFT OUTER` joined to `gold_dim_ts` (`ts_id`) and `tbl_cb_sdv_raw_tr_v2` (`tr_id`).

4. **TR ↔ Issue Bridge:** `tbl_cb_sdv_raw_tr_v2` → (`LEFT OUTER`, 1:N) → `fact_testrun_issue` → (`LEFT OUTER`, N:1) → `tbl_cb_sdv_raw_issue_v2`.

- **Information Needed:** Confirmation of the Explore base view, whether unlinked Test Cases/Test Runs must be retained (`FULL OUTER`) or filtered out (`LEFT OUTER` from Requirements), and the exact join condition on `FACT_TC_TR_TS_REQ`.

2.3 Modeling Strategy for BigQuery `ARRAY<...>` Columns

- **Observation:** Many key attributes across `gold_dim_req`, `gold_dim_ts`, `gold_dim_fact_req_tc`, and the Raw tables are repeated fields (`ARRAY<STRING>`, `ARRAY<NUMERIC>`, `ARRAY<INT64>`), such as `Req_MySF`, `Req_Baseline`, `Req_Allocation`, `Req_Regulation`, and `TS_MySF`.
- **Current Assumption:** Both the raw array field and a flattened display dimension using `ARRAY_TO_STRING(field, ', ')` were created so users can filter via `contains` without causing row fan-out.
- **Information Needed:** Which `ARRAY` columns require individual element breakdown in charts (requiring `LEFT JOIN UNNEST(...)` virtual views in the Explore) vs. comma-separated string display.

2.4 Business KPIs, Status Lifecycles & Verdict Rules

- **Observation:** The diagram specifies `measures` and `business_metrics` at a conceptual level without SQL formulas or CodeBeamer status values.
- **Current Assumption:**
  - **Requirement Coverage Rate (%):** Distinct `Req_Id` present in `gold_dim_fact_req_tc` divided by total distinct `Req_Id` in `gold_dim_req`.
  - **Accepted/Validated Statuses:** `UPPER(status) IN ('ACCEPTED', 'APPROVED', 'RELEASED', 'VALIDATED')`.
  - **Test Run Verdicts:** `Passed` (`PASSED/PASS/OK`), `Failed` (`FAILED/FAIL/KO/NOK`), `Blocked` (`BLOCKED`).
  - **Test Run Pass Rate (%):** `Passed Test Runs / Executed Test Runs` (where Executed = Passed + Failed + Blocked).
  - **Open Issues:** Issues where `UPPER(Status) NOT IN ('CLOSED', 'RESOLVED', 'DONE', 'REJECTED', 'CANCELLED', 'VERIFIED')`.
- **Information Needed:** Exact KPI definitions and the list of distinct CodeBeamer status/result values used in production.

2.5 Derived Attributes (`TR_MySF`) & Field Curation on 114-Column Tables

- **Observation:**
  1. In the diagram, `FACT_TC_TR_TS_REQ` sends `TR_MySF` to `TestRun_View`. Physically, `TR_MySF` is a column in `FACT_TC_TR_TS_REQ`.
  2. `TBL_CB_SDV_Raw_TC_V2` and `TBL_CB_SDV_Raw_TR_V2` each contain 114 columns, including many technical/tracker fields.
- **Current Assumption:** `tr_my_sf` is defined in `fact_tc_tr_ts_req`, and all 114 columns in the TC and TR tables are currently exposed.
- **Information Needed:** Whether `tr_my_sf` should be visually grouped into the **Test Run** folder in Looker (via `view_label`), and whether a subset of the 114 raw TC/TR columns should be hidden (`hidden: yes`) to simplify the self-service UI.

2.6 Target BigQuery & Looker Environment Configuration

- **Observation:** BigQuery dataset separation (Gold vs. Raw/Silver) and Looker connection names were not specified.
- **Current Assumption:** Parameterized via constants in `manifest.lkml` (`CONNECTION_NAME = "bigquery_ampere_sdv"`, `DATASET_NAME = "ampere_sdv_gold"`).
- **Information Needed:** Whether Gold tables (`gold_dim_*`, `FACT_*`) and Raw tables (`TBL_CB_SDV_Raw_*`) reside in the same BigQuery dataset or two separate datasets (e.g., `GOLD_DATASET_NAME` and `RAW_DATASET_NAME`).

3. Customer Input Checklist (To Complete)

To finalize the LookML project with zero assumptions, please provide the details in the checklist below:

#	Requirement Item	Required Input / Question	Customer Response
1	`FACT_TestRun_Issue` **Schema**	Exact physical table name, column names, and data types in BigQuery.	
2	**BigQuery Dataset Names**	Do Gold tables (`gold_dim_*`, `FACT_*`) and Raw tables (`TBL_CB_SDV_Raw_*`) live in one dataset or two separate datasets? Provide exact dataset name(s).	
3	**Looker Connection Name**	Exact Looker connection name for the model `ampere_sdv_analytics`.	
4	**Explore Join Rules**	Confirm base view (`gold_dim_req`), join types (`LEFT` vs. `FULL OUTER` for orphan TCs/TRs), and join keys on `FACT_TC_TR_TS_REQ`.	
5	**Array Unnesting**	List any `ARRAY<...>` columns that must be unnested via `LEFT JOIN UNNEST` for row-by-row grouping in Looker.	
6	**CodeBeamer Status & Verdict Values**	Provide exact production values for: • `Req_Status` / `Req_FinalStatus` • `TC.Status` & `TC.Automated` • `TR.Result` & `TR.Status` • `Issue.Status` & `Issue.Resolution`	
7	**Official KPI Formulas**	Confirm exact numerator & denominator rules for: • Requirement Coverage Rate (%) • Test Case Automation Rate (%) • Test Run Execution & Pass Rate (%) • Defect Resolution Rate (%)	
8	**Field Visibility (114-col Raw Tables)**	Should all 114 columns in `TBL_CB_SDV_Raw_TC_V2` and `TBL_CB_SDV_Raw_TR_V2` remain visible, or should only key business fields be exposed?	
9	**Cache & Datagroup Policy**	Expected BigQuery ETL refresh frequency and preferred SQL trigger column.	